

1. What is Hashing?

Definition: Hashing is a process of mapping data of arbitrary size to fixed-size values, usually for faster data retrieval.

Key Characteristics:

- **Deterministic:** The same input will always produce the exact same hash.
 - **Fast:** Designed to be highly efficient for data retrieval tasks.
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2. Why Use Hashing?

- **Efficient Data Retrieval:** Allows for near-instant access to data using a specific key.
 - **Hash Tables:** This is the most common data structure that utilizes hashing to enable fast lookups.
 - **Applications:** Widely used in databases, caches, cryptography, and more.
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3. How Hashing Works

The process follows a specific three-step flow:

1. **Overview:** Raw data is processed through a **hash function** to generate a unique hash code.
 2. **Hash Code:** This is the fixed-size representation of your original input data.
 3. **Hash Table:** The data is then stored in a table, using that hash code as the specific **index** for its location.
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4. Hash Functions & Properties

A hash function takes an input (like a file or a name) and produces a fixed-size string of characters (the hash value).

Core Properties:

- **Deterministic:** Same input yields the same hash every time.
 - **Efficient:** The value is computed quickly.
 - **Uniform Distribution:** Hash values should be spread out evenly across the table to keep the system organized.
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5. Collision Resolution

What is a Collision?

A collision occurs when two different inputs accidentally produce the exact same hash code.

Techniques to Fix This:

- **Chaining:** Using linked lists at each index so multiple items can stay at the same spot.
- **Open Addressing:** Finding the next available empty slot in the table if the first one is taken.

6. Common Applications

- **Databases:** Quick data retrieval using indexes.
- **Caches:** Storing frequently accessed data so it loads faster.
- **Cryptography:** Using hash functions for data integrity verification (checking if a file has been changed).
- **Password Storage:** Hashing passwords so they are stored securely rather than in plain text.